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Handling Multiple Windows and Frames

MAY, 2026



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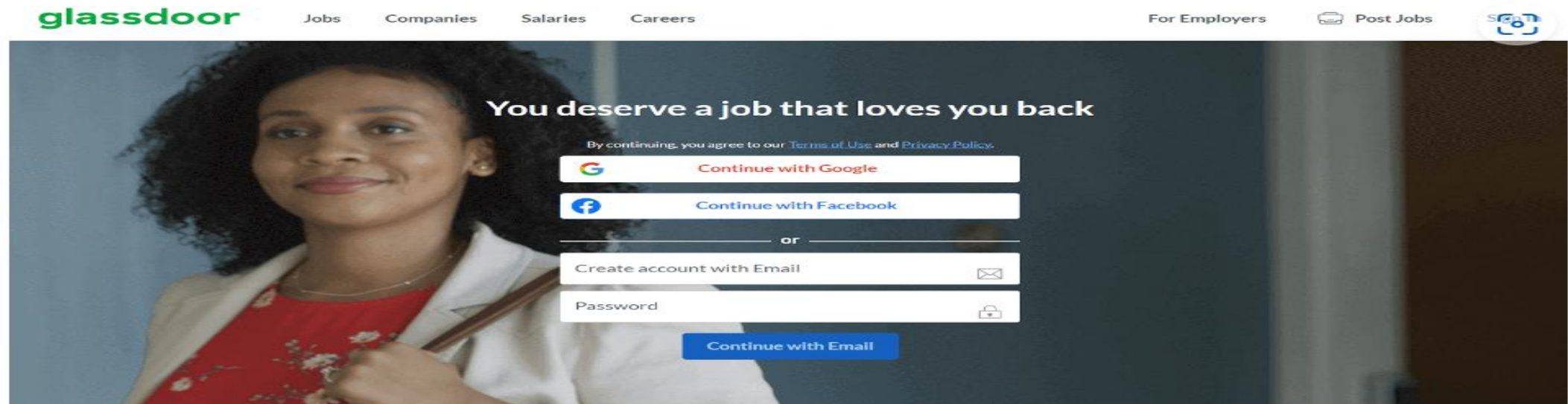
Windows in Selenium



Windows in Selenium

Introduction

- In Selenium Automation Testing, when we talk about “windows”, we’re referring to the different browser windows or tabs that can open up when you interact with a web application.
- These windows can be the primary browser window or ones that arise after clicking links or buttons.
- For example, the job search platform Glassdoor opens a new window when you click the buttons to sign in using your Google account or Facebook account.

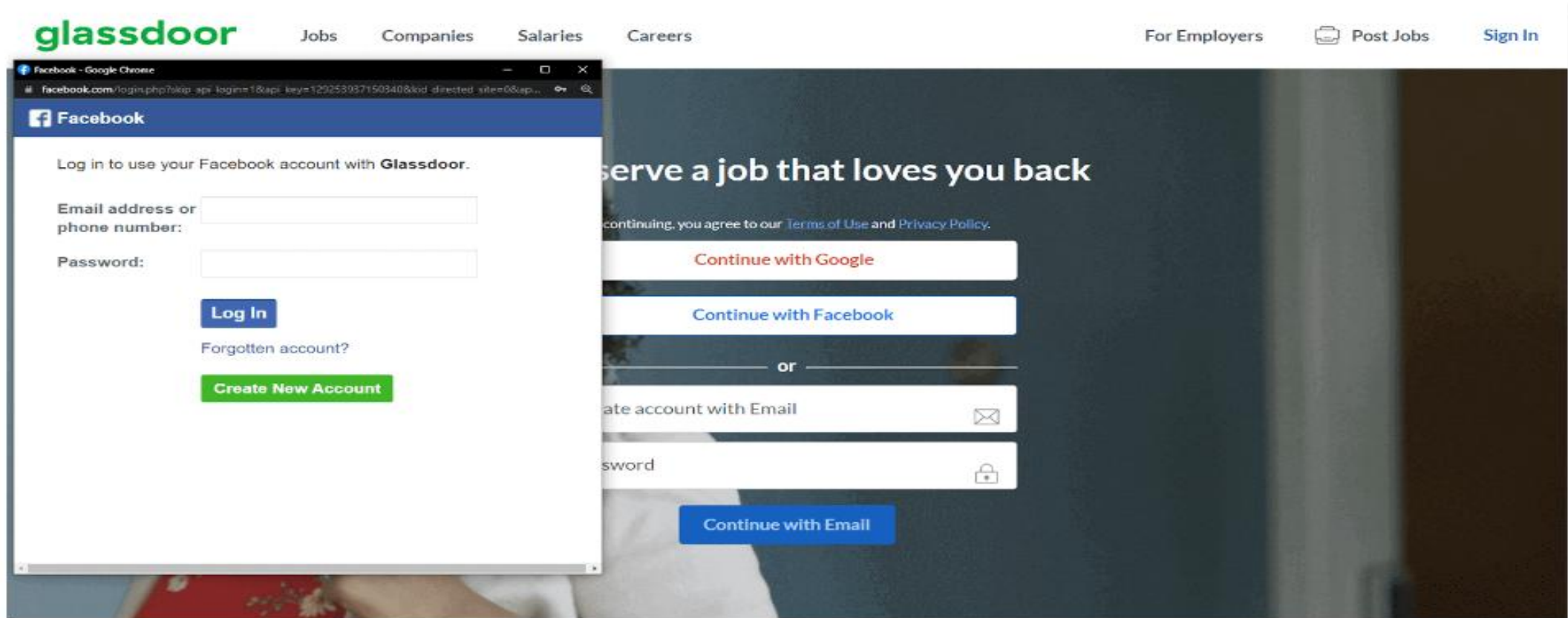


Glassdoor Home Page

Windows in Selenium

Introduction

- Click on the “Continue with Facebook” button to sign in.



Facebook Page Window

Windows in Selenium

Introduction

- A Facebook webpage window will pop up to prompt you to sign in.
- As a result, the user might need to access the open browser window to move forward.
- As a QA tester, running a test case might require you to complete some operations in a newly opened browser window or tab before going back to the original window or tab to finish the rest of the tasks.
- There are two types of Windows In Selenium
 1. Main Window
 2. Child Window

Windows in Selenium

Introduction

1. Main Window

- The main window is the initial browser window that opens when you start a Selenium test. It's like the front door of a house.
- You interact with the elements on this window as you navigate through the web application.
- **For Example:**
- Let's say you're testing an online shopping website.
- The main window is where you land when you open the website.
- You can click on different products, add them to your cart, and navigate through various pages – all within this main window.

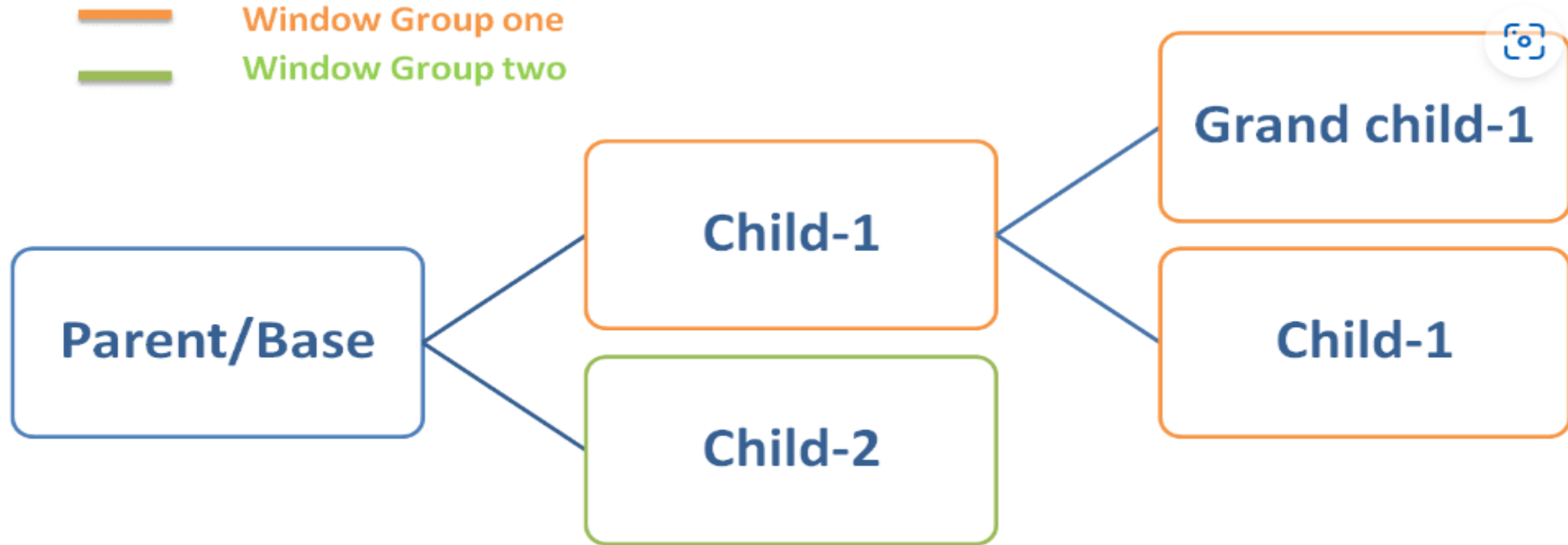
Introduction

2. Child Window

- Child windows, on the other hand, are additional windows that pop up when you perform certain actions.
- These could be like opening new doors within the house that lead to different rooms.
- **For Example:**
- Back to the online shopping example, imagine you're looking at a product and you click a "Details" button.
- A new window might pop up with more information about the product. This is a child window.
- You can interact with elements in this window, like reading the product description, without losing track of the main window.

Windows in Selenium

Introduction



The Hierarchy of Browser Windows

Windows Handle and Methods



Windows Handle and Methods

Window Handle

- Multiple windows are brought up by scenarios in which a web application navigates to or opens multiple browser windows.
- Selenium provides the option to handle multiple windows with window handles.
- A **window handle** is like an **address for browser windows**.
- It's a **unique code** made of letters and numbers that **helps Selenium identify and manage windows**.
- Just as a key opens a specific room in a building, a window handle lets Selenium switch between different windows.
- Each window has its own handle, making it easy to work with multiple windows and their child windows while testing a website..

Methods To Handle Windows In Selenium

1) **current_window_handle()**: This method returns the unique handle of the currently focused window. It's often used to identify the main or parent window.

Syntax: *driver.current_window_handle*



```
1 driver.get('https://www.lambdatest.com/selenium-playground')
2 ....
3 ....
4 driver.current_window_handle
```

Methods To Handle Windows In Selenium

2) `window_handles()`: This method retrieves the handles of all open windows. It returns a set of strings, each representing a unique window handle.

Syntax: *`driver.window_handles`*



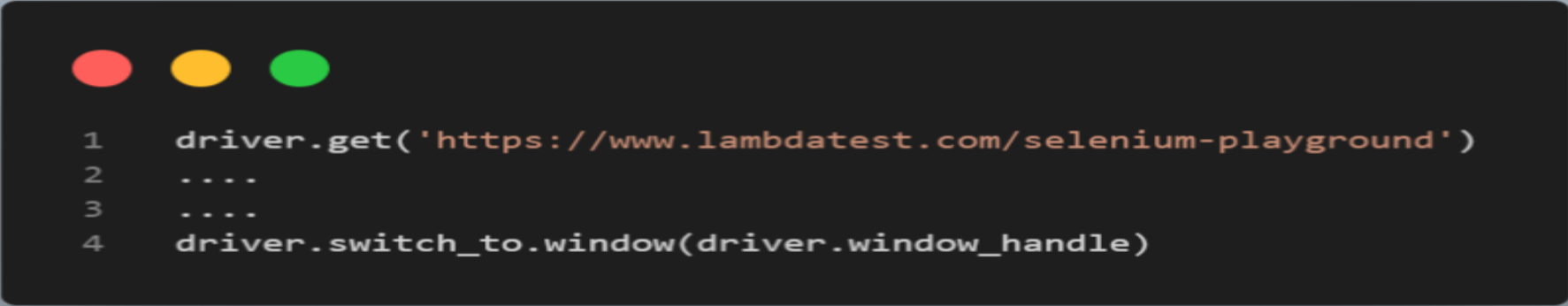
```
1  driver.get('https://www.lambdatest.com/selenium-playground')
2  ....
3  ....
4  driver.window_handles
5
```

Methods To Handle Windows In Selenium

3) `switch_to.window()`: This method allows you to switch your focus between different windows. You provide the window handle as an argument, and Selenium switches your interaction context to that window.

Syntax:

Option 1: `driver.switch_to.window()`



```
1 driver.get('https://www.lambdatest.com/selenium-playground')
2 ....
3 ....
4 driver.switch_to.window(driver.window_handle)
```

How to handle multiple windows



How to handle multiple windows

step by step guide on how to handle multiple windows

Step 1: Open Main Window

from selenium import webdriver



```
# Create a new browser window (main window)
driver = webdriver.Chrome()
driver.get("https://www.example.com")
```

How to handle multiple windows

step by step guide on how to handle multiple windows

Step 2: Trigger Action for New Window



```
# Click a button to open a new window (simulated)
open_help_button = driver.find_element_by_id("open-help-button")
open_help_button.click()
```

How to handle multiple windows

step by step guide on how to handle multiple windows

Step 3: Capture Window Handles



```
# Capture handles of all open windows  
main_window_handle = driver.current_window_handle  
all_window_handles = driver.window_handles
```

How to handle multiple windows

step by step guide on how to handle multiple windows

Step 4: Switch to New Window



```
# Switch to the new window
for window_handle in all_window_handles:
    if window_handle != main_window_handle:
        driver.switch_to.window(window_handle)
        break
```

How to handle multiple windows

step by step guide on how to handle multiple windows

Step 5: Do Stuff in the New Window



```
# Interact with elements in the new window (simulated)
name_input = driver.find_element_by_id("name-input")
name_input.send_keys("John Doe")
```

How to handle multiple windows

step by step guide on how to handle multiple windows

Step 6: Switch Back to Main Window



```
# Switch back to the main window  
driver.switch_to.window(main_window_handle)
```

How to handle multiple windows

step by step guide on how to handle multiple windows

Step 7: Continue Testing on Main Window



```
# Interact with elements in the main window
main_button = driver.find_element_by_id("main-button")
main_button.click()
```

How to handle multiple windows

step by step guide on how to handle multiple windows

Step 8: Repeat as Needed



```
# Open another new window (simulated)
open_another_window_button = driver.find_element_by_id("open-another-window-button")
open_another_window_button.click() # Capture handles and switch to the new window
# Interact with elements, switch back, etc.
```

How to handle multiple windows

step by step guide on how to handle multiple windows

Step 9: Close or Quit



```
# Close the current window  
driver.close() # Quit the entire browser session  
driver.quit()
```

Handling Multiple Windows - Demo

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()

driver.get("https://omayo.blogspot.com/")

parent_windowID = driver.current_window_handle

#Accessing child window problem
driver.find_element(By.LINK_TEXT, value="Open a popup window").click()

time.sleep(5)

window_text = driver.find_element(By.XPATH, value="//h3[text()='New Window']").text
print(window_text)

time.sleep(5)

driver.quit()
```

Output

Output:

```
[Running] python -u "c:\Users\Arun Kumar\VSCodePython\multiplewindows.py"
Traceback (most recent call last):
  File "c:\Users\Arun Kumar\VSCodePython\multiplewindows.py", line 18, in <module>
    para_text = driver.find_element(By.ID, value="para1").text
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Arun Kumar\AppData\Local\Programs\Python\Python312\Lib\site-packages\selenium\webdriver\remote\webdriver.py", line 742, in find_element
    return self.execute(Command.FIND_ELEMENT, {"using": by, "value": value})["value"]
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Arun Kumar\AppData\Local\Programs\Python\Python312\Lib\site-packages\selenium\webdriver\remote\webdriver.py", line 348, in execute
    self.error_handler.check_response(response)
  File "C:\Users\Arun Kumar\AppData\Local\Programs\Python\Python312\Lib\site-packages\selenium\webdriver\remote\errorhandler.py", line 229, in check_response
    raise exception_class(message, screen, stacktrace)
selenium.common.exceptions.NoSuchElementException: Message: no such element: Unable to locate element: {"method":"css selector","selector":["id="para1"]}
(Session info: chrome=121.0.6167.85); For documentation on this error, please visit: https://www.selenium.dev/documentation/webdriver/troubleshooting/errors#no-such-element-exception
Stacktrace:
   GetHandleVerifier [0x00007FF7AEA85E42+3538674]
   (No symbol) [0x00007EE7AE6A4C02]
```

Handling Multiple Windows - Demo

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://omayo.blogspot.com/")

parent_windowID = driver.current_window_handle

#Handling Multiple Windows
driver.find_element(By.LINK_TEXT, value="Open a popup window").click()
windowsID = driver.window_handles

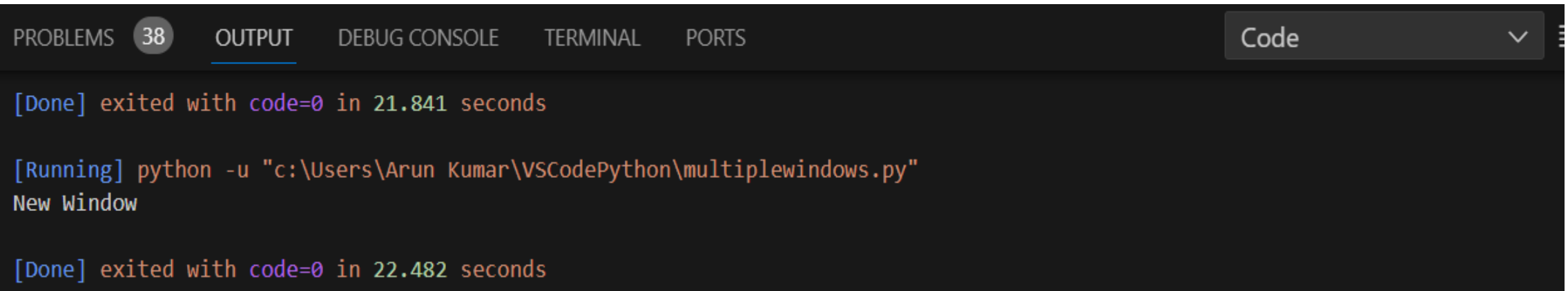
for w in windowsID:
    driver.switch_to.window(w)
    if driver.title.__eq__("New Window"):
        window_text = driver.find_element(By.XPATH, value="//h3[text()='New Window']").text
        print(window_text)
        driver.close()
        break
time.sleep(5)
```

Handling Multiple Windows - Demo

```
driver.switch_to.window(parent_windowID)
driver.find_element(By.ID, value="ta1").send_keys("Hello Welcome")

time.sleep(5)

driver.quit()
```

A screenshot of a VS Code terminal window. The top bar shows tabs for PROBLEMS (38), OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The OUTPUT tab is active, displaying the following text: [Done] exited with code=0 in 21.841 seconds, [Running] python -u "c:\Users\Arun Kumar\VSCodePython\multiplewindows.py", New Window, and [Done] exited with code=0 in 22.482 seconds. A 'Code' button with a dropdown arrow is visible in the top right corner of the terminal panel.

```
PROBLEMS 38 OUTPUT DEBUG CONSOLE TERMINAL PORTS Code
```

```
[Done] exited with code=0 in 21.841 seconds
```

```
[Running] python -u "c:\Users\Arun Kumar\VSCodePython\multiplewindows.py"
```

```
New Window
```

```
[Done] exited with code=0 in 22.482 seconds
```

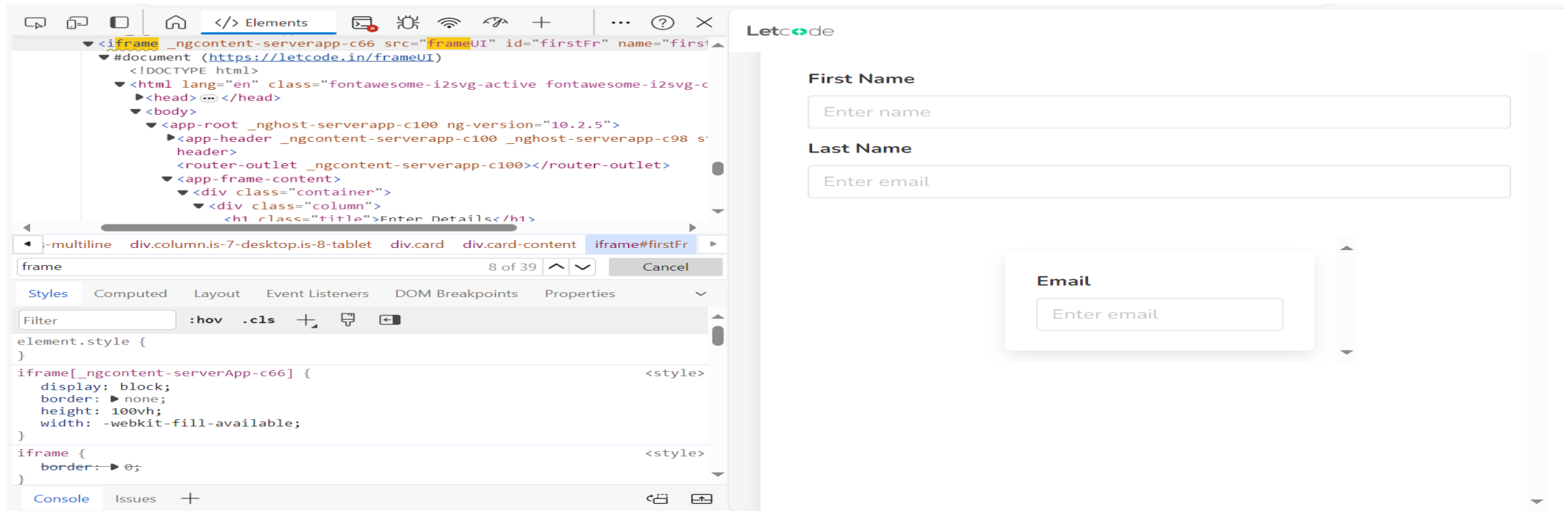
Handling Frames



Handling Frames

iframes

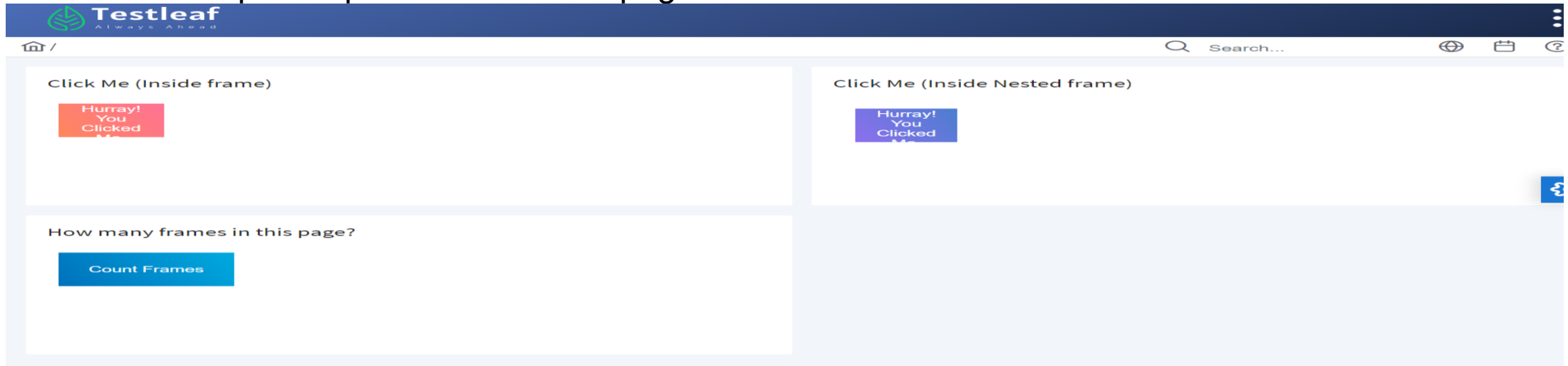
- An iframe is also known as an inline frame. It is a tag used in HTML5 to embed an HTML document within a parent HTML document.
- An iframe tag is defined using `<iframe></iframe>` tags



Handling Frames

frame vs iframe

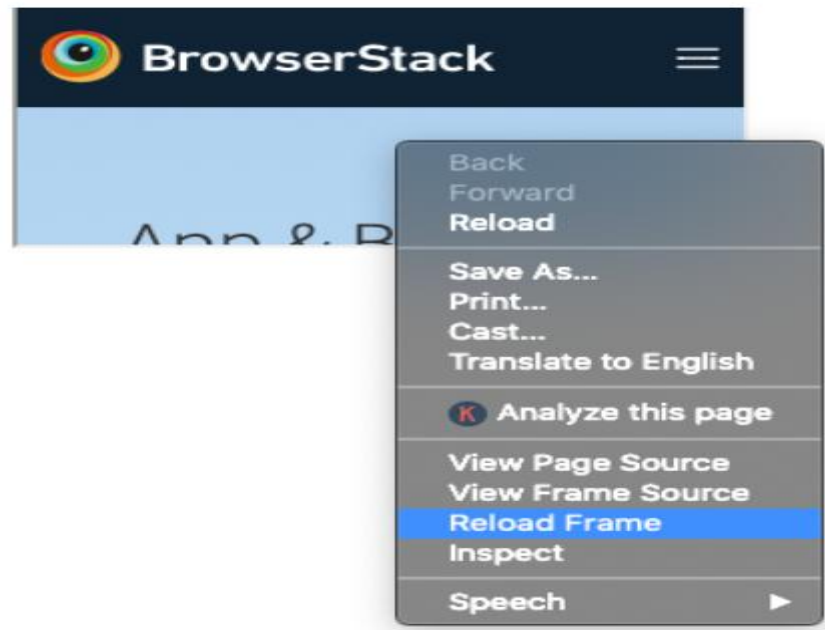
- The **frame** enables a developer to split the screen horizontally or vertically by using the frameset tag.
- **Note:** Frame and frameset tags are deprecated as they are no longer supported by HTML 5.
- The iframes are mainly used to insert content from external sources. For example, an advertisement displayed on a web page. They can float within the webpage, which means one can position an iframe at a specific position on a web page.



Handling Frames

How to identify a frame on a page?

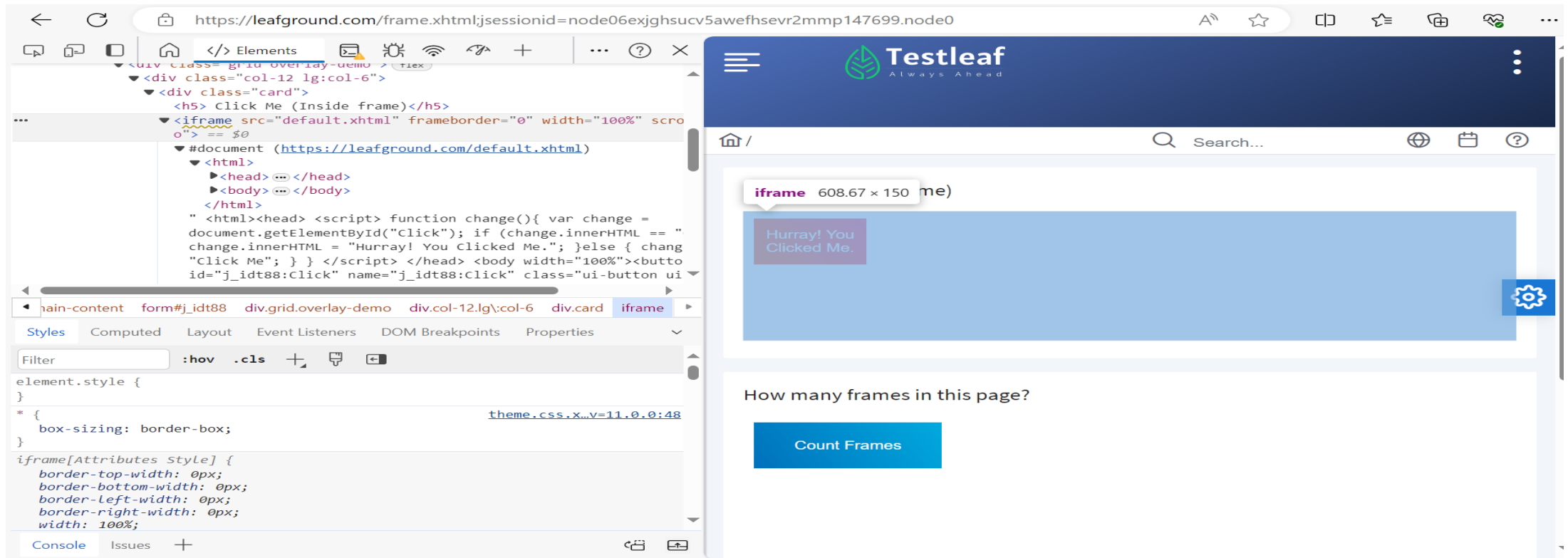
- It is possible to identify the iframes on a web page in two ways:
- Right-click on the specific element and check all the options. If you find an option like **This Frame, view Frame source or Reload Frame**, the page includes frames.
- Consider the image below as an example.



Handling Frames

How to identify a frame on a page?

- Similar to the first step, right-click on the page and click on View Page Source. On the page source, **search for “iframe-tags”**. If you find any iframe tags, it means the page includes iframes.



Handling Frames

Using the `switch_to().frame` function

- This method enables the browser to switch between multiple frames. It can be implemented in the following ways:

1. `switch_to.frame(int frame number)`:

- Defining the frame index number, the Driver will switch to that specific frame

Example:

```
driver.switchTo().frame(0); // Index 0 for example
```

Handling Frames

Using the `switch_to().frame` function

2. `switch_to.frame(string frameNameOrId)`:

- Defining the frame Name or Id, the Driver will switch to that specific frame.

Example:

```
self.driver.switch_to.frame(frame1)
```

3. `switch_to.frame(WebElement frameElement)`:

- Defining the frame web element, the Driver will switch to that specific frame.

Example:

```
iframe = self.driver.find_element(By.XPATH, "//iframe")  
self.driver.switch_to.frame(iframe)
```

Handling Frames

Using the `switch_to().frame` function

4. `driver._switch_to.parent_frame()`

- switches the control to the parent frame of the current frame.

5. `driver._switch_to.default_content()`

- switches the control to the main web page regardless of the number of frames.

Example:

- Suppose there are three frames named i1,i2 and i3 within the parent web page p1. Frames i1, i2, and i3 are dependant on each other which means one frame will be the parent of another.
- Using `driver.switchTo().defaultContent()` method on frame i3, web driver control moves to the parent page, p1. While `driver.switchTo().parentFrame()` method on frame i3 switches the control back to frame i2 and so on.

Handling Frames - Demo

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()
# Handling Frames
driver.get("https://letcode.in/frame")

time.sleep(5)

driver.switch_to.frame("firstFr")
driver.find_element(By.NAME, value="fname").send_keys("Arunkumar")
driver.find_element(By.NAME, value="lname").send_keys("Arumugam")
time.sleep(5)

child_frame = driver.find_element(By.XPATH, value="//iframe[@class='has-background-white']")
driver.switch_to.frame(child_frame)
driver.find_element(By.NAME, value="email").send_keys("aarun.83@gmail.com")

time.sleep(5)
```

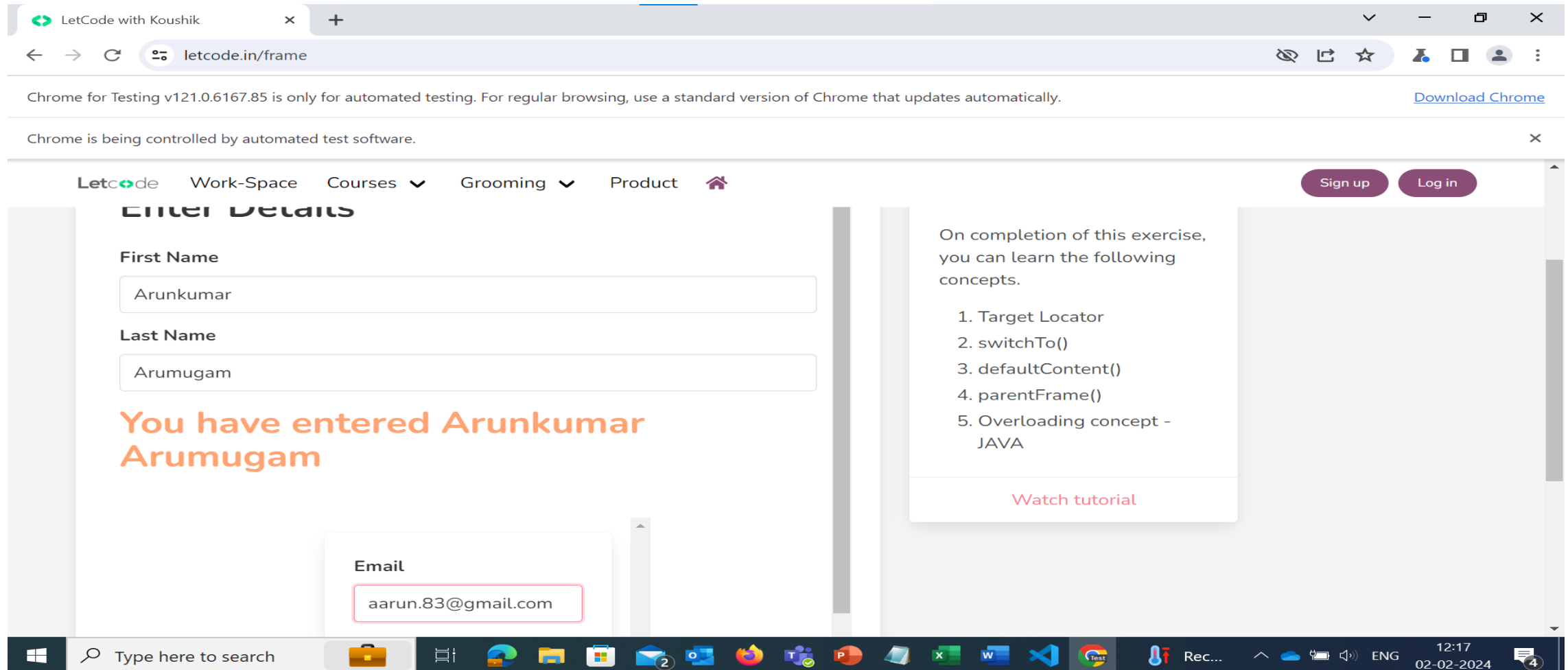
Handling Frames - Demo

```
""" driver.switch_to.parent_frame()
driver.find_element(By.NAME, value="fname").clear()
"""

driver.switch_to.default_content()
driver.find_element(By.XPATH, value="//a[@id='testing']").click()

time.sleep(5)
driver.quit()
```

Handling Multiple Windows - Demo



LetCode with Koushik

letcode.in/frame

Chrome for Testing v121.0.6167.85 is only for automated testing. For regular browsing, use a standard version of Chrome that updates automatically. [Download Chrome](#)

Chrome is being controlled by automated test software.

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Sign up Log in

Enter Details

First Name

Arunkumar

Last Name

Arumugam

You have entered Arunkumar Arumugam

Email

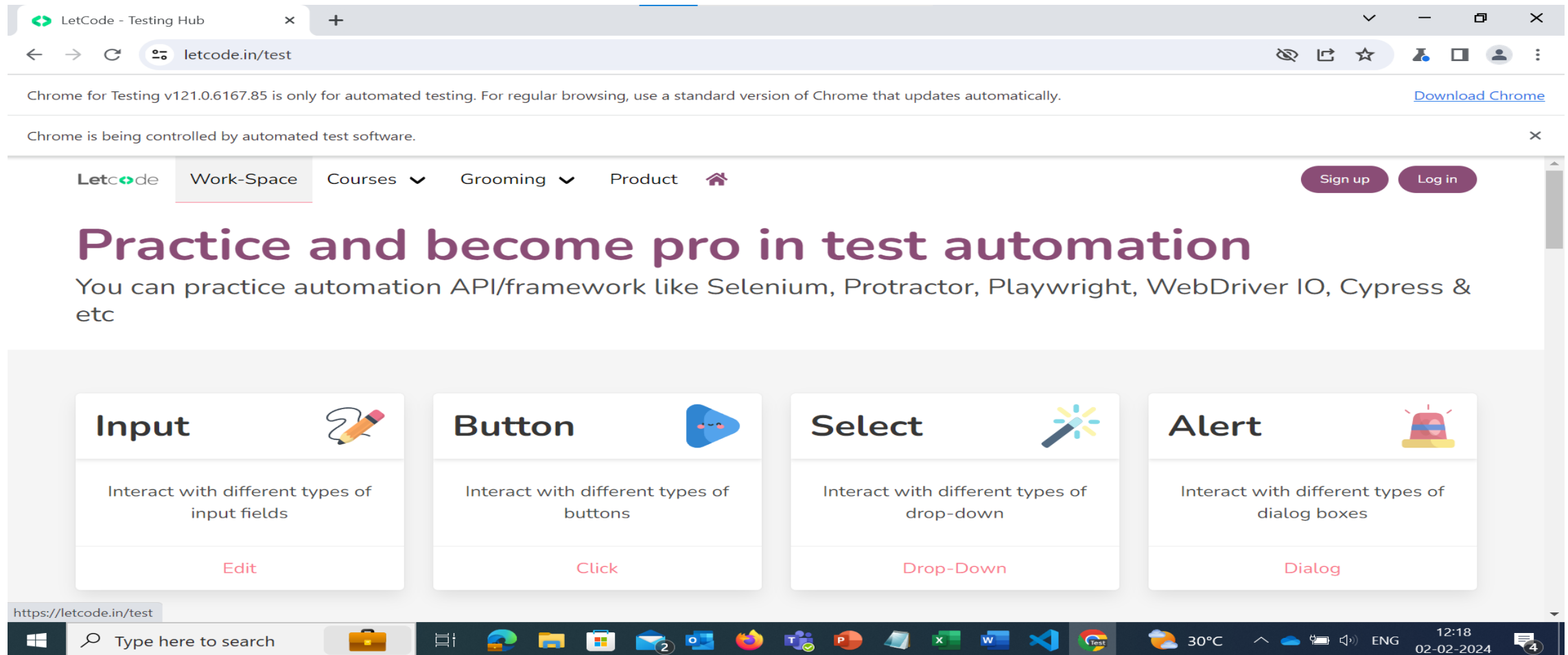
aarun.83@gmail.com

On completion of this exercise, you can learn the following concepts.

1. Target Locator
2. switchTo()
3. defaultContent()
4. parentFrame()
5. Overloading concept - JAVA

Watch tutorial

Handling Multiple Windows - Demo



LetCode - Testing Hub

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Practice and become pro in test automation

You can practice automation API/framework like Selenium, Protractor, Playwright, WebDriver IO, Cypress & etc

Input

Interact with different types of input fields

Edit

Button

Interact with different types of buttons

Click

Select

Interact with different types of drop-down

Drop-Down

Alert

Interact with different types of dialog boxes

Dialog

https://letcode.in/test

Type here to search

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THANK YOU